

Patient Multi-omics

- Gene Expression
- Mutated genes
- Mutation types
- Clinical Traits
- BeatAML train: waves 1-2
- BeatAML test: waves 3-4



Gene expression
Mutated Genes
Mutation Types
Clinical Traits



Drugs

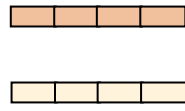
- Physicochemical features (PC)
- Embedding representation (Embed)



Sources:

BeatAML drugs, DrugBank, MOSES, ChEMBL, ChemBERTa, MolFormer-XL

RDkit → PC
Transfer /
— → Embed
Zero-Shot
learning



Existing Data

- Pathway Database
- String PPI network
- BeatAML (Cell & Module states)



Pathway Genesets
PPI + drug targets
Cell + Module
Genesets



Pathway
enrichments

Cell + Module
Enrichment

Drug Pathway
AUC

Data

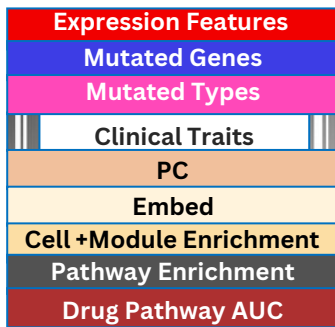
Feature engineering

Modeling

Analysis

Featured used

Models



Linear Models:

- Ridge Regression
- Linear SVM

Tree Models:

- Random Forests
- XGBoost
- LightGBM
- Catboost

DeepLearning Models:

- FFNN
- CNN + DNN
- LSTM + DNN
- TabPFN

$AUC \sim f(p, d, k)$

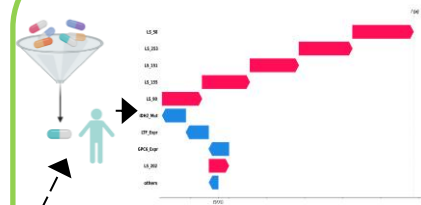
Cell Viability
Drug Conc.

Stratification

Random
Patient
Drug

Best
Model
Drug

SHAP Explanations



Patient Stratification



External Validation



LeeAML
FIMM-AML